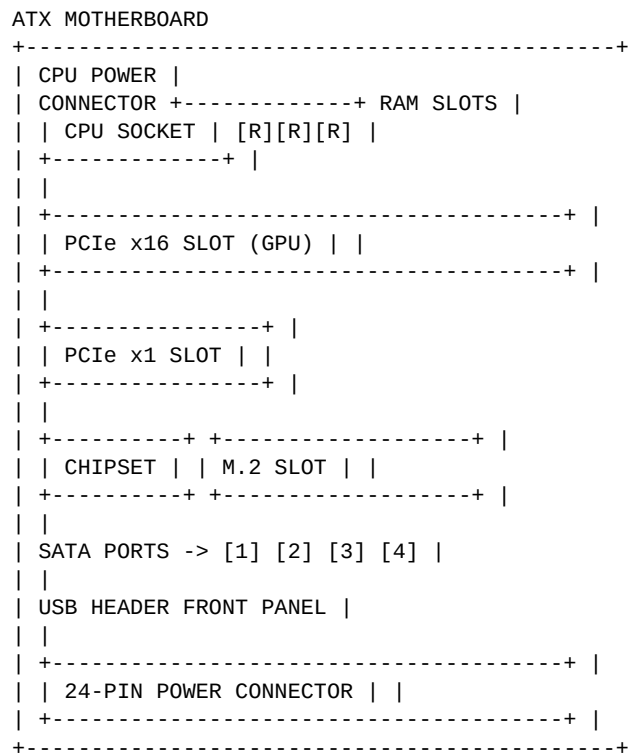


Day 2: Motherboard and CPU

Easy-to-Learn Theory + Practical Assignment Answers

1. ATX Motherboard — Labeled Diagram



Part	Easy Meaning
CPU Socket	Place where the CPU is installed.
RAM / DIMM Slots	Slots where RAM memory is installed.
PCIe x16 Slot	Main expansion slot commonly used for a graphics card.
PCIe x1 Slot	Smaller expansion slot for compatible add-in cards.
SATA Ports	Connect SATA storage devices such as HDDs and SATA SSDs.
M.2 Slot	Slot commonly used for an M.2 SSD.
USB Header	Connects compatible front-panel USB ports.
24-pin Power Connector	Main power connection from the power supply to the motherboard.
CPU Power Connector	Provides power to the CPU area.
Chipset	Helps manage communication between the CPU and many motherboard-connected devices.

2. Popular CPU Socket Types

CPU Socket	Compatible CPU Example
LGA1151	Intel Core i7-9700
AM4	AMD Ryzen 5 5600

CPU Socket	Compatible CPU Example
LGA1200	Intel Core i5-10400F

Easy memory: LGA1151 → i7-9700 | AM4 → Ryzen 5 5600 | LGA1200 → i5-10400F

Important: Socket matching alone is not always enough. The motherboard chipset and BIOS can also affect CPU compatibility.

3. CPU and Chipset Functions

CPU – Central Processing Unit

The CPU is the main processor of the computer. It performs instructions and calculations needed to run applications.

Chipset

The chipset helps manage communication between the CPU and many motherboard-connected devices, such as USB, storage and other expansion features.

Example: Opening Spotify

You click Spotify → the CPU processes the instructions → the chipset and motherboard handle communication with connected hardware → RAM holds the data currently needed → Spotify opens.

Easy example:

Computer Part	Like in an Office
CPU	Manager — does the main thinking/work.
Chipset	Coordinator — helps different parts communicate.
RAM	Desk — keeps things currently being worked on.

4. Real Motherboard Photo — What to Identify

For the practical assignment, use a clear real motherboard photo and circle/label at least five of these parts:

- ① CPU Socket
- ② RAM Slots
- ③ PCIe x16 Slot
- ④ SATA Ports
- ⑤ 24-pin Power Connector
- ⑥ USB Header
- ⑦ CPU Power Connector
- ⑧ M.2 Slot
- ⑨ Chipset
- ⑩ Front Panel Header

■ Very Easy Revision

Motherboard → Main board

CPU → Processes instructions

RAM → Temporary memory

Chipset → Helps components communicate

PCIe → Expansion / GPU slot

SATA → Storage connection

M.2 → Common SSD slot

USB Header → Internal USB connection

24-pin → Main motherboard power